

Sunav Sunil Mattoo

Software Engineer | Backend Development | Data Structures & Algorithms

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Professional Summary

Third-year B.Tech (Artificial Intelligence and Machine Learning) student and full-stack/backend developer with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, Software Engineering, and Database Management Systems. Skilled in designing and building scalable backend and full-stack applications using Java, Spring Boot, Apache Kafka, Redis, PostgreSQL, React, and REST APIs, with hands-on exposure to event-driven architecture, asynchronous processing, distributed systems, and real-time applications. Solved 800+ Data Structures and Algorithms problems (peak LeetCode Contest Rating: 1692) across competitive programming platforms. Seeking Software Development Engineer (SDE) Intern and AI/ML Intern roles.

Education

VIT Bhopal University

2024 – 2028

Bachelor of Technology (B.Tech), Artificial Intelligence and Machine Learning

CGPA: 8.0 / 10.0

Technical Skills

Programming Languages: Java, C++, Python, JavaScript, SQL

Software Engineering: Data Structures and Algorithms, Object-Oriented Programming (OOP), Object-Oriented Design, Design Patterns, Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Development Life Cycle (SDLC), Problem Solving, Debugging and Troubleshooting

Backend & Web Development: Spring Boot, REST API Development, FastAPI, React.js, Next.js, Node.js, Express.js, WebSocket, STOMP

Distributed Systems: Apache Kafka, Redis, Event-Driven Architecture, Asynchronous Processing, Fault-Tolerant Systems, Distributed Computing, Real-Time Systems, Microservices Architecture

Databases, Cloud & Tools: SQL, NoSQL, PostgreSQL, MongoDB, Firebase, Firestore, AWS (Academic Exposure), Git, GitHub, Postman, Visual Studio Code, Version Control

Artificial Intelligence & Machine Learning: Generative AI, Large Language Model (LLM) Integration, AI-Powered Development Tools, Machine Learning, Pandas, NumPy, Scikit-learn, TensorFlow, XGBoost, Prophet

Projects

CodeAscend AI — Competitive Programming and AI-Assisted Evaluation Platform

- Built a competitive programming platform using Java 21, Spring Boot, React, Apache Kafka, Redis, and PostgreSQL, applying Object-Oriented Programming, layered architecture, software engineering principles, and RESTful backend design.
- Architected a fault-tolerant, event-driven distributed system using Apache Kafka for asynchronous, non-blocking submission evaluation and a Redis ZSET-based leaderboard supporting ranked queries in $O(\log N)$ time.
- Implemented an Elo rating engine with anti-farming safeguards and integrated LLM-powered agents for automated code review, algorithmic complexity analysis, and non-spoiler hints; version-controlled using Git and GitHub.

VITHub — AI-Powered Campus Digital Twin

- Developed a real-time campus platform using Next.js 16, Spring Boot 3.5 (Java 21), and PostgreSQL, implementing layered backend services across 16 normalized relational tables with Flyway-managed database migrations.
- Engineered a WebSocket (STOMP) communication pipeline achieving sub-100ms state synchronization for live occupancy tracking, shuttle GPS/ETA updates, and library seat availability.
- Implemented JWT authentication, role-based access control (RBAC), REST APIs, and Swagger/OpenAPI documentation, following maintainable software development and backend design practices.

FinSight AI — AI-Powered Personal Finance Intelligence Platform

- Developed a full-stack financial platform using Next.js, FastAPI, Firebase, and REST APIs for real-time expense tracking, budget management, and transaction analytics.
- Built analytics services using Pandas, Scikit-learn, and Prophet for forecasting and anomaly detection; implemented Firebase Authentication, Firestore, and an LLM-powered financial copilot.

Achievements

- Solved 800+ Data Structures and Algorithms problems across LeetCode, CodeChef, GeeksforGeeks, and other competitive programming platforms.
- Solved 470+ problems on LeetCode and achieved a peak LeetCode Contest Rating of 1692 across numerous contests, alongside a CodeChef Rating of 1556 and a DSA Rating of 1651, reflecting a strong passion for competitive programming.
- Secured 2nd Rank in a college-wide Data Structures and Algorithms competition among 100+ participating teams.

Certifications

- Oracle AI Foundations Associate
- JPMorgan Chase Software Engineering Job Simulation
- HP LIFE Data Science and Analytics